2068 Interface

File: interface.kicad_sch

Local Bus and Registers

File: memory.kicad_sch

RP2350B Core

File: mcu.kicad_sch

Power

File: power.kicad_sch

SD and Display

File: storage_display.kicad_sch

WiFi

File: wifi.kicad_sch

Audio

File: audio.kicad_sch

Joystick

File: joystick.kicad_sch

Root sheet: see docs/v3-architecture.md

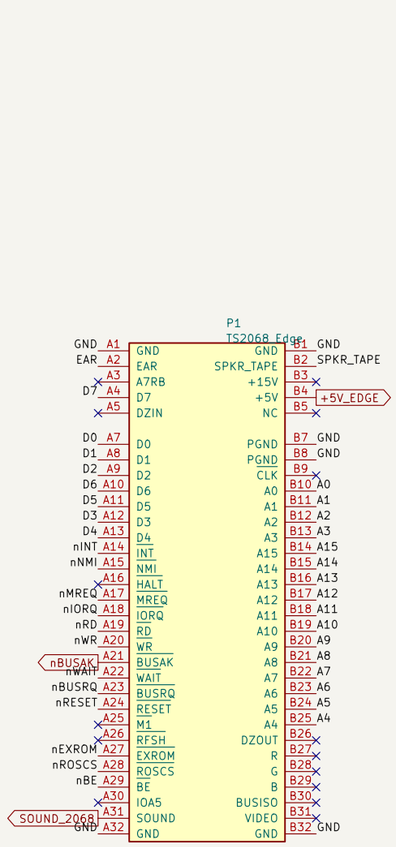
Timex/Sinclair Online User Group

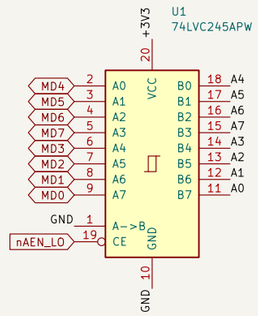
Sheet: /

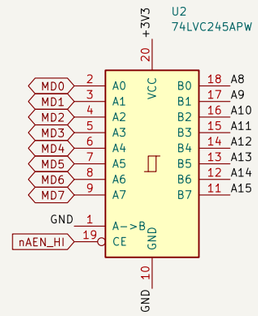
File: tspico-v3.kicad_sch

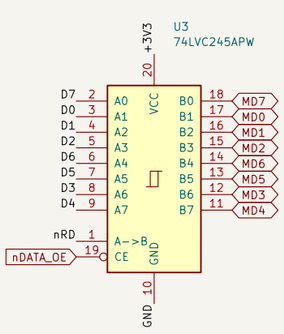
Title: TS-Pico v3

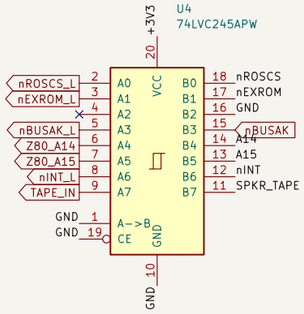
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 1/9

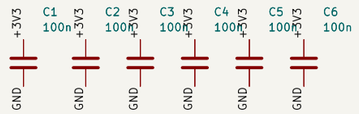


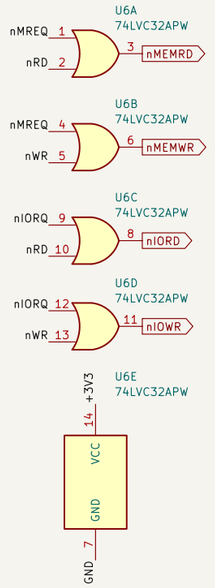


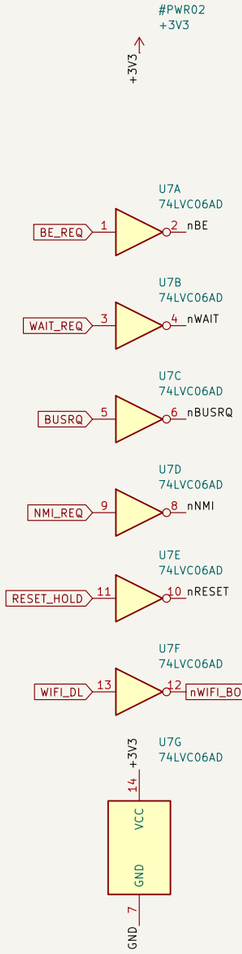


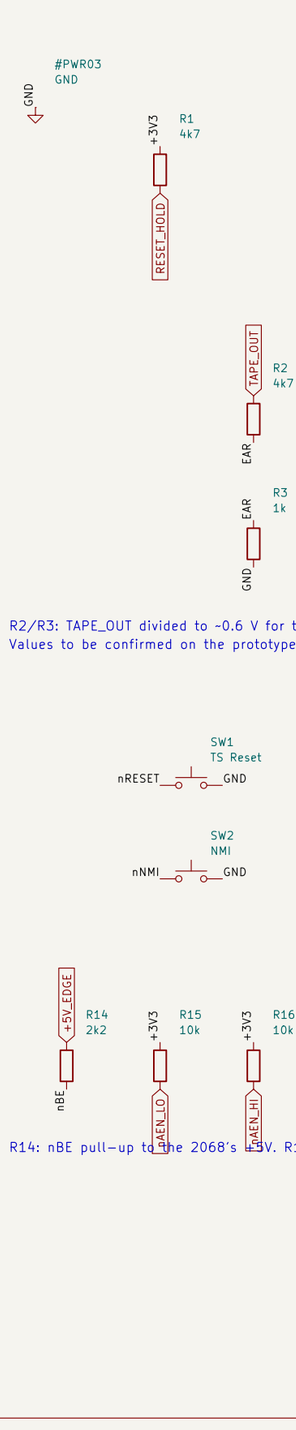








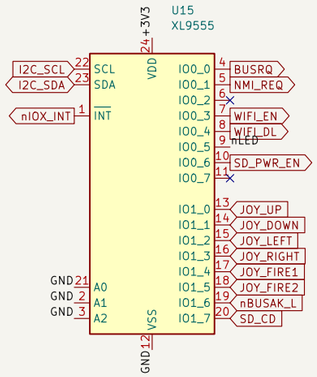


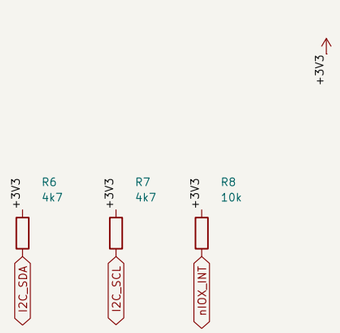


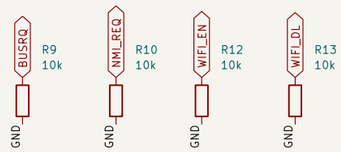
TS2068 bus interface. Nets: n = active low, _L = 3.3 V copy of a bus signal, _G = 3.3 V request driven onto the bus through an open-drain buffer.

Power: edge +5V (B4) goes to the power sheet as +5V_EDGE and is diode-ORed with USB VBUS. +15V (B3) unused.

Timex/Sinclair Online User Group	
Sheet: /2068 Interface/	
File: interface.kicad_sch	
Title: TS-Pico v3: 2068 Interface	
Size: A3 / Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5	Id: 2/9







R9, R10, R12, R13: 10 k pull-downs beat the expander's 100 k pull-ups until the firmware configures the port.

R5/D4: activity LED, on when nLED is driven low.



Slow control and status on one I2C expander. The fast local bus MD0–MD7 (address groups, Z80 data) is on the interface sheet.

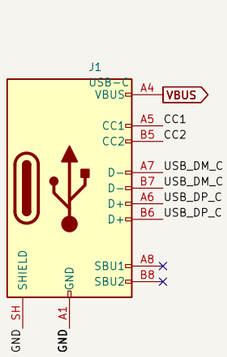
No flash or SRAM: the RP2350B serves memory from internal SRAM (PIO + DMA).

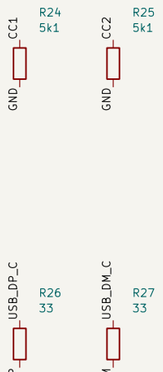
Timex/Sinclair Online User Group

Sheet: /Local Bus and Registers/
File: memory.kicad_sch

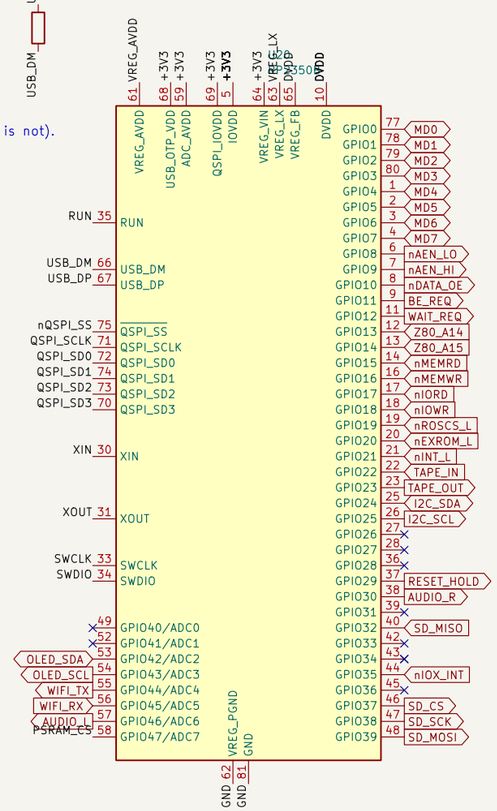
Title: TS-Pico v3: I/O Expander

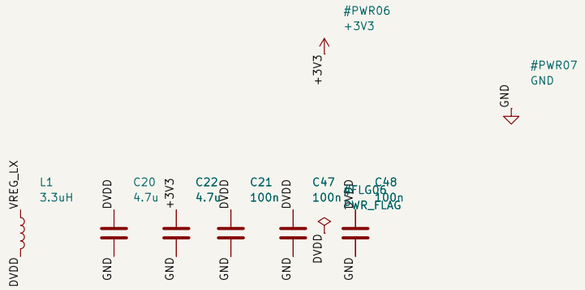
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.5		Id: 3/9



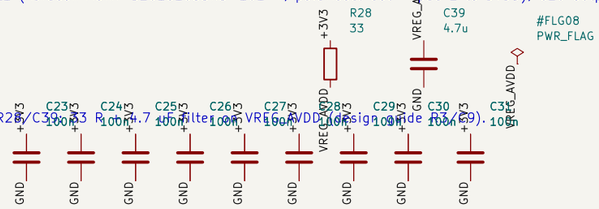


J1: USB 2.0. VBUS -> power sheet. SBU unconnected.
R24/R25 5.1 k Rd. R26/R27 33 R series (JLC basic; 27 R is not).



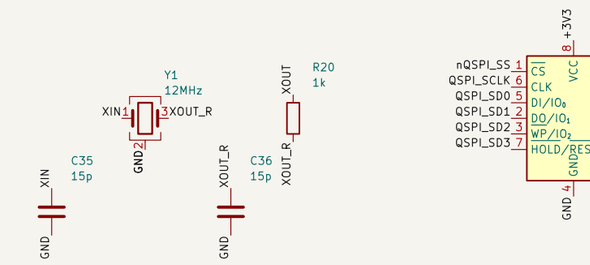


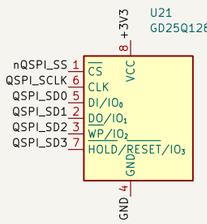
L1 (Abracon AOTA-B201610S3R3-101-T, polarised: dot = pad 2 to DVDD), C20 output, C22 at VREG_VIN (0402 as in the design guide). C21/C47/C4



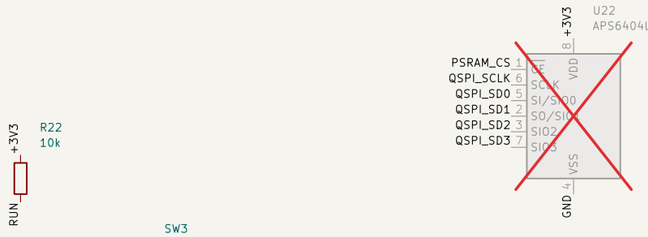
R28/C39 33 R + 4.7 µF ferrite from VREG_AVDD (design guide 9.4).

C23–C31: one per 3.3 V supply pin or adjacent pair (5, 15, 24, 29, 41, 50, 59/60, 68/69, 76); pin 64 has C22.

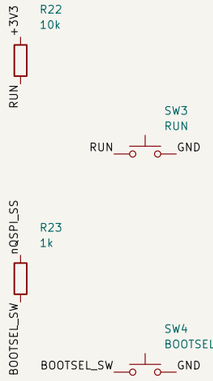




Y1: 12 MHz, e.g. ABM8-272-T3 (CL 10 pF). C35/C36 15 pF, R20 1 k in series with XOUT.



U21 boot flash (XIP CS0). U22 PSRAM on the same QSPI bus, DNP (fit APS6404L-3S
CS1 = GPIO47 (QMI CS1n). C37/C38 at U21/U22.





TP1–TP3: SWD debug pads.

RP2350B core per the RP2350 hardware design guide. GPIO map: docs/v3-architecture.md section 7. GP34, GP36, GP40, GP41 spare (plus the DMA pins when the DMA option is off).

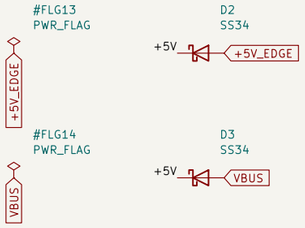
USB-C: data, and VBUS to the power sheet (diode-ORed with the edge +5V) so the card runs from USB alone on the bench.

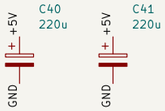
Timex/Sinclair Online User Group	
Sheet: /RP2350B Core/	
File: mcu.kicad_sch	
Title: TS-Pico v3: RP2350B Core	
Size: A3 / KiCad E.D.A. 10.0.5	Date:
Rev: 0.1	Id: 4/9



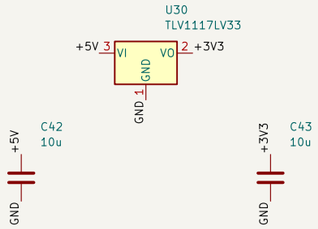




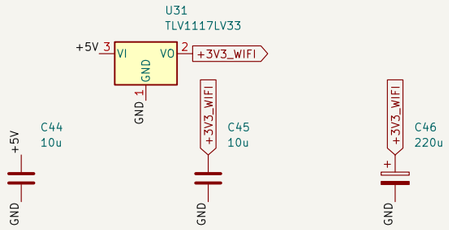




C40/C41: bulk for WiFi transmit bursts (SMD aluminium electrolytic).



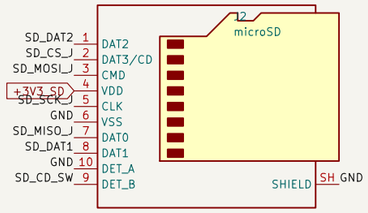
D2/D3: SS34 (3 A, SMA). About 0.35 V drop at 0.5 A; the 1117-LV LDOs need only 0.35 V headroom.



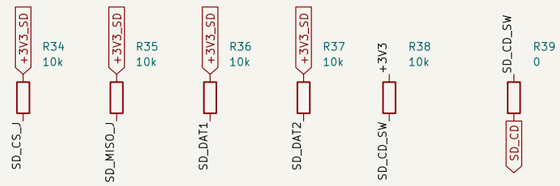
U30/U31: TLV1117LV33 (SOT-223, 1 A, 0.35 V dropout, 1117 pinout). Alternate: AMS1117-3.3 (1.1 V dropout, marginal).
Input current budget: about 250 mA average, 600 mA peak with WiFi transmit. Do not exceed 1 A from the 2068's own rail.

+5V rail = diode-OR of the edge connector +5V (B4) and USB VBUS. Card runs from the 2068/expansion bus, from USB alone, or both.
Two LDOs: +3V3 (RP2350B, logic, SD, OLED) and +3V3_WIFI (ESP32-C3 module, 350 mA transmit bursts).

Timex/Sinclair Online User Group		
Sheet: /Power/		
File: power.kicad_sch		
Title: TS-Pico v3: Power		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 5/9

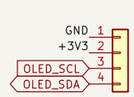








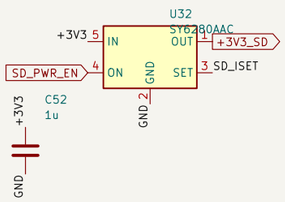


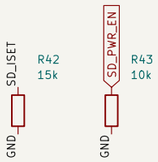


J3 pin order matches the module: GND VCC SCL SDA. VCC is +3V3, so the module's own pull-ups stay at 3.3 V.



R30–R33 33 R series; R34–R37 10 k pull-ups (CS, MISO, DAT1, DAT2) on +3V3_SD; R38 pull-up on the detect switch (+3V3), R39 0 R link to SD_CD.
Card present = SD_CD low (the TF-01A switch closes to GND; confirmed on proto1). U32: SD VDD switch, R42 sets 0.45 A, R43 holds it off.
Firmware order: card detected, settle, SD_PWR_EN on, then SPI. SPI lines released (inputs) before SD_PWR_EN goes off.

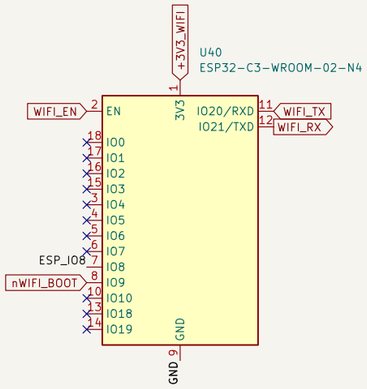


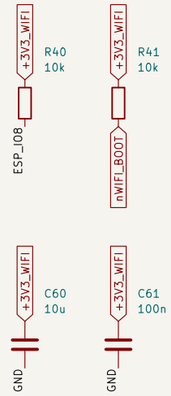


Micro SD in SPI mode on the RP2350B SPI0 (GP36–GP39); card detect (switch to GND) into STAT bit 7.

OLED: 4-pin header for a 1.54in 128x64 SSD1309 I2C module on I2C1 (GP42 SDA, GP43 SCL), its own bus so display updates never delay joystick reads.

Timex/Sinclair Online User Group		
Sheet: /SD and Display/		
File: storage_display.kicad_sch		
Title: TS-Pico v3: SD and Display		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 6/9







R40: IO8 must be high during a download-mode boot. IO9 low = download mode: R41 pulls it up, the open-drain gate in U7 pulls it down when WIFI_DL is set.
Zimodem's modem-status pins (DCD, RI, DSR, DTR, RTS/CTS) are not routed; XON/XOFF flow control.

ESP32-C3-WROOM-02 running Zimodem (Hayes modem over UART). Antenna end over the board edge, copper keep-out per the module datasheet.
Flashing: RP2350B acts as USB-UART bridge; firmware sets WIFI_DL=1 then pulses WIFI_EN for download mode.

Timex/Sinclair Online User Group		
Sheet: /WiFi/ File: wifi.kicad_sch		
Title: TS-Pico v3: WiFi		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 7/9



C70
1u



R70
100k



R71
1k



C71
10n



R72
1k



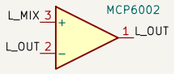
C72
10n



R73
10k



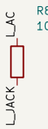
R74
22k



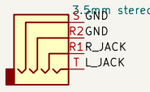
U50A
MCP6002



C80
10u



R80
100





R75
1k



C75
10n



R76
1k



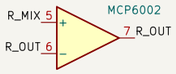
C76
10n



R77
10k



R78
22k



U50B
MCP6002



C81
10u



R81
100

U50C
MCP6002



C79
100n

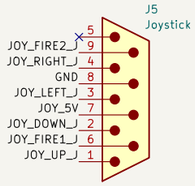




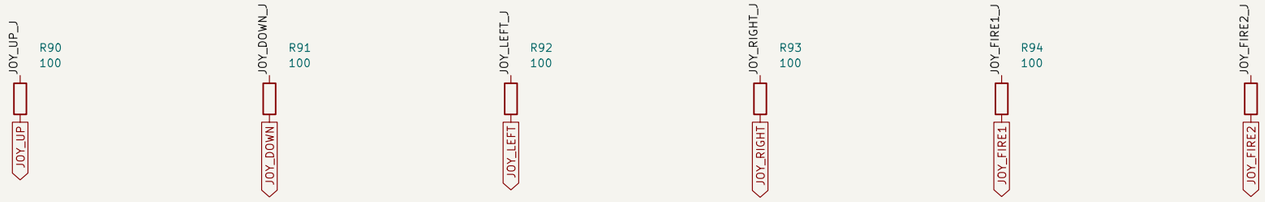
Mix: L = AY A + AY B/2, R = AY C + AY B/2 done in firmware; SOUND_2068 added to both channels here.
The PWM branch (0 to 3.3 V, DC about 1.65 V) also biases the buffer input; SOUND is AC-coupled by C70.

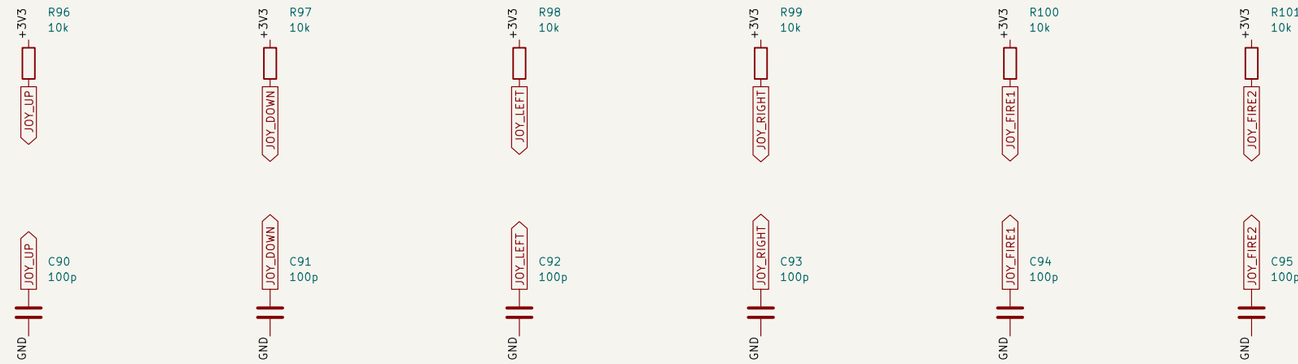
Emulated AY-3-8910 on RP2350B PWM (AUDIO_L/R), low-passed, mixed with the 2068 SOUND line, buffered to a stereo 3.5 mm jack.
Op-amp on the +5V rail (rail-to-rail MCP6002). Component values are starting points; tune on the prototype.

Timex/Sinclair Online User Group	
Sheet: /Audio/	
File: audio.kicad_sch	
Title: TS-Pico v3: Audio	
Size: A3 / Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5	Id: 8/9









Kempston joystick on a male DE-9 (Atari pinout). Inputs pulled up to +3V3, read through STAT; firmware serves port $1F (bit = 1 when pressed).

JP10: optional +5V on pin 7 for autofire joysticks (solder jumper, open by default).

Timex/Sinclair Online User Group		
Sheet: /Joystick/		
File: joystick.kicad_sch		
Title: TS-Pico v3: Joystick		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 9/9